COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	8/27/26	**DATE SUBMITTED:**	8/29/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issued this command. Did something happen? Why?

```
villareal1@workstation:~/CPE232_villareal$ git pull
Already up to date.
```

- Displayed a message that says “Already up to date” indicating that the updated version of the remote repository has been integrated to the local branch.

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): ansible-playbook --ask-become-pass install_apache.yml. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
villareal1@workstation:~/CPE232_villareal$ ansible-playbook -i inventory.yaml ins
tall_apache.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [ubuntu1]
ok: [ubuntu2]
fatal: [centos1]: FAILED! => {"msg": "Incorrect sudo password"}

TASK [update repository index] *************************************************
changed: [ubuntu1]
changed: [ubuntu2]

TASK [install apache2 package] *************************************************
ok: [ubuntu1]
changed: [ubuntu2]

TASK [add PHP support for apache] **********************************************
ok: [ubuntu1]
changed: [ubuntu2]

PLAY RECAP *********************************************************************
centos1                    : ok=0    changed=0    unreachable=0    failed=1    sk
ipped=0    rescued=0    ignored=0
ubuntu1                    : ok=4    changed=1    unreachable=0    failed=0    sk
ipped=0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=3    unreachable=0    failed=0    sk
ipped=0    rescued=0    ignored=0
```

3. Edit the install_apache.yml file and insert the lines shown below. Make sure to save the file and exit. Run ansible-playbook --ask-become-pass install_apache.yml and describe the result. If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

```
- name: update repository index
  apt:
     update_cache: yes
  when: ansible_distribution in ["Debian", "Ubuntu]
```

Note: This will work also if you try. Notice the changes are highlighted.

```
-ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [centos]
ok: [node1]
ok: [node2]

TASK [update repository index] *************************************************
skipping: [centos]
changed: [node1]
changed: [node2]

TASK [install apache2 package] *************************************************
skipping: [centos]
ok: [node2]
ok: [node1]

TASK [add PHP support for apache] **********************************************
skipping: [centos]
ok: [node2]
ok: [node1]

PLAY RECAP *********************************************************************
centos                     : ok=1    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
node1                      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node2                      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

- Instead of the centos failing it instead skips.

4. Edit the install_apache.yml file and insert the lines shown below. Make sure to save and exit. Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [centos]
ok: [node2]
ok: [node1]

TASK [update repository index] *************************************************
skipping: [centos]
changed: [node2]
changed: [node1]

TASK [install apache2 package] *************************************************
skipping: [centos]
ok: [node1]
ok: [node2]

TASK [add PHP support for apache] **********************************************
skipping: [centos]
ok: [node1]
ok: [node2]

TASK [update repository index] *************************************************
skipping: [node1]
skipping: [node2]
ok: [centos]

TASK [install apache2 package] *************************************************
skipping: [node1]
skipping: [node2]
changed: [centos]

TASK [add PHP support for apache] **********************************************
skipping: [node1]
skipping: [node2]
changed: [centos]

PLAY RECAP *********************************************************************
centos                     : ok=4    changed=2    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
node1                      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
node2                      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

- There is now a separate part within the playbook that is built for CentOS.

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in CentOS is not yet active. Thus, you need to activate it first.

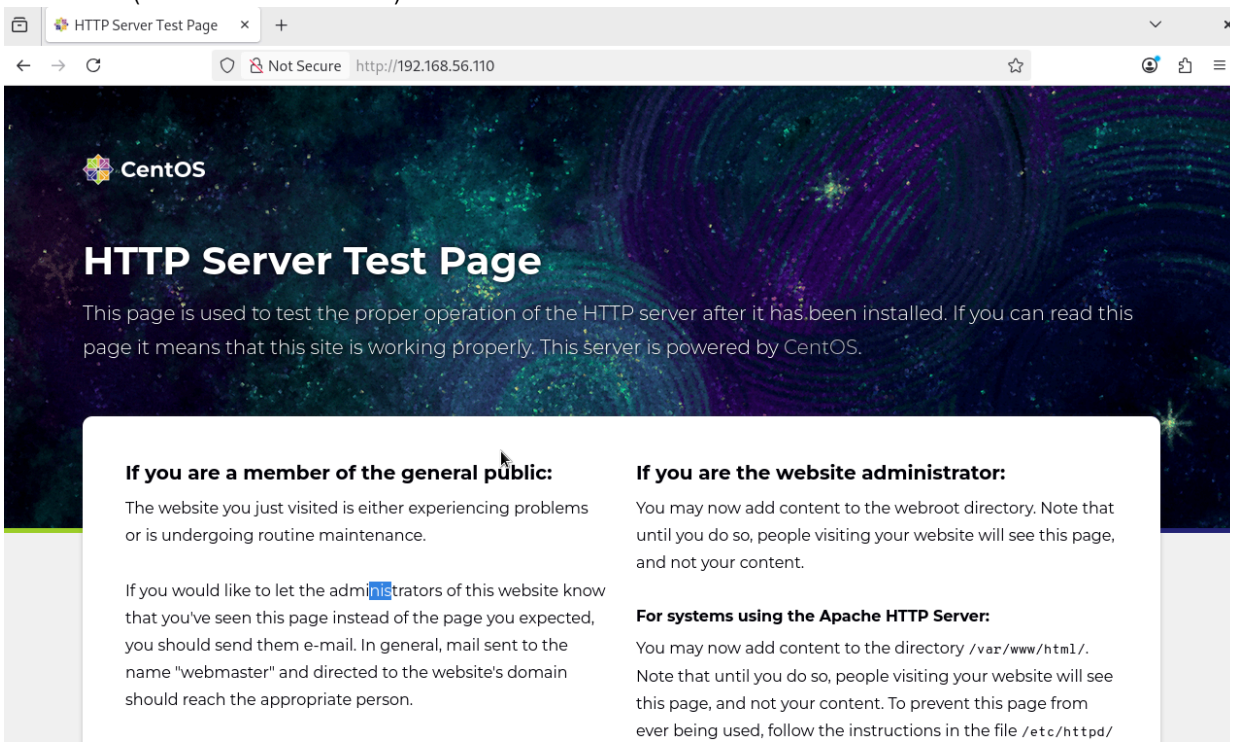
5.1 To activate, go to the CentOS VM terminal and enter the following: systemctl status httpd The result of this command tells you that the service is inactive.

```
[villareal@localhost ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: d>
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead)
       Docs: man:httpd.service(8)
```

5.2 Issue the following command to start the service: sudo systemctl start httpd (When prompted, enter the sudo password) sudo firewall-cmd --add-port=80/tcp (The result should be a success)

```
[villareal@localhost ~]$ sudo systemctl start httpd
[sudo] password for villareal:
[villareal@localhost ~]$ sudo firewall-cmd --add-port=80/tcp
success
```

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also make run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook install_apache.yml. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the

same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
ok: [node1]
ok: [node2]

TASK [update repository index Ubuntu] ***************************************
skipping: [centos]
changed: [node1]
changed: [node2]

TASK [install apache2 and php package for Ubuntu] ****************************
skipping: [centos]
ok: [node2]
ok: [node1]

TASK [update repository index for CentOS] ************************************
skipping: [node1]
skipping: [node2]
ok: [centos]

TASK [install apache and php packages for CentOS] ****************************
skipping: [node1]
skipping: [node2]
ok: [centos]

PLAY RECAP *******************************************************************
centos                     : ok=3    changed=0    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
node1                      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
node2                      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
```

- Same output but shorter playbook commands.

2. Edit the playbook install_apache.yml again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidate everything in just 2 plays. This can be done by removing the update repository play and putting the command update_cache: yes below the command state: latest. See below for reference:

Make sure to save the file and exit.
Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory -
-ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [node2]
ok: [node1]
ok: [centos]

TASK [install apache2 and php package for Ubuntu] ******************************
skipping: [centos]
ok: [node1]
ok: [node2]

TASK [install apache and php packages for CentOS] ******************************
skipping: [node1]
skipping: [node2]
ok: [centos]

PLAY RECAP *********************************************************************
centos                     : ok=2    changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
node1                      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
node2                      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
```

- Even less playbook common lines for minimal output.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook install_apache.yml again and make sure to follow the below image.

   Make sure to save the file and exit.
   Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory -
-ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [centos]
ok: [node1]
ok: [node2]

TASK [install apache and php] **************************************************
fatal: [node1]: FAILED! => {"msg": "The task includes an option with an undefine
d variable. The error was: 'apache_package' is undefined. 'apache_package' is un
defined\n\nThe error appears to be in '/home/villareal/CPE212_villareal/install_
apache.yml': line 6, column 5, but may\nbe elsewhere in the file depending on th
e exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: inst
all apache and php\n    ^ here\n"}
fatal: [node2]: FAILED! => {"msg": "The task includes an option with an undefine
d variable. The error was: 'apache_package' is undefined. 'apache_package' is un
defined\n\nThe error appears to be in '/home/villareal/CPE212_villareal/install_
apache.yml': line 6, column 5, but may\nbe elsewhere in the file depending on th
e exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: inst
all apache and php\n    ^ here\n"}
fatal: [centos]: FAILED! => {"msg": "The task includes an option with an undefin
ed variable. The error was: 'apache_package' is undefined. 'apache_package' is u
ndefined\n\nThe error appears to be in '/home/villareal/CPE212_villareal/install
_apache.yml': line 6, column 5, but may\nbe elsewhere in the file depending on t
he exact syntax problem.\n\nThe offending line appears to be:\n\n\n  - name: ins
tall apache and php\n    ^ here\n"}
```

- It shows an unsuccessful execution of the playbook since “apache_package” is undefined.

4. Unfortunately, task 2.3 was not successful. It’s because we need to change something in the inventory file so that the variables we declared will be in place. Edit the inventory file and follow the below configuration:

   Make sure to save the inventory file and exit.

Finally, we still have one more thing to change in our install_apache.yml file. In task 2.3, you may notice that the package is assigned as apt, which will not run in CentOS. Replace the apt with the package. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use apt, and for CentOS it will automatically use dnf. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: ansible.builtin.package – Generic OS package manager — Ansible Documentation

Run ansible-playbook --ask-become-pass install_apache.yml and describe the result.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory -
-ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [node2]
ok: [node1]
ok: [centos]

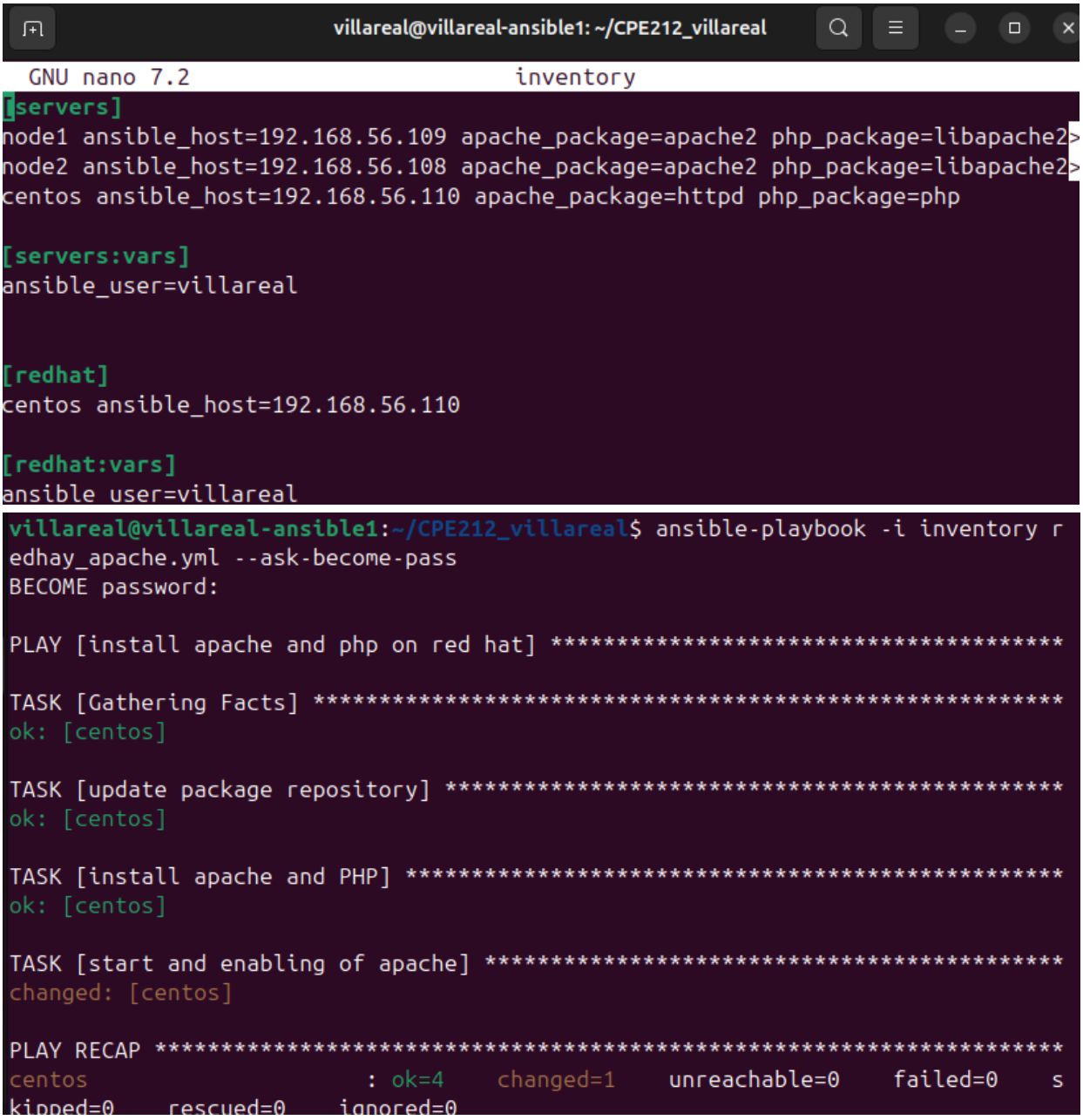
TASK [install apache and php] **************************************************
ok: [centos]
ok: [node2]
ok: [node1]

PLAY RECAP *********************************************************************
centos                     : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node1                      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
node2                      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

- The playbook now runs without problems since the issue is that there is no definition on what “apache_package” and “php_package” but since there are changes to the inventory where both the commands are defined it works as intended.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Create a playbook that could do the previous tasks in Red Hat OS.



Reflections:
Answer the following:

1. Why do you think refactoring of playbook codes is important?
- I think refactoring is important because it makes the playbook simpler, cleaner, and easier to understand. It can also reduce repeated tasks and make Ansible run more efficiently.
2. When do we use the “when” command in the playbook?
- We use the when command when we want a task to run only if a certain condition is true. For example, we used it to make specific tasks run only on Ubuntu or CentOS depending on the operating system.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

COMPUTER ENGINEERING

QUEZON CITY

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error encounter and Solving of certain issues**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to help resolve issues unrelated to the activity, for example: CentOS VM's second adapter is not active therefore AI is used for setting up the VM.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.